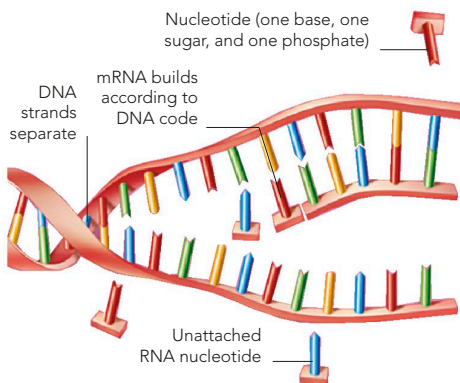


HOW DNA WORKS

One of DNA's key functions is to provide the information to build proteins. Some proteins are the body's major structural molecules, while others form enzymes or hormones, which control chemical reactions within the body. Manufacture of proteins occurs in two main phases: transcription and translation. In transcription, information is taken from the DNA and copied to an intermediate type of molecule called mRNA (messenger ribonucleic acid). The mRNA moves out of the cell's

nucleus to protein assembly units called ribosomes. In the translation phase, the mRNA acts as a template for the formation of units of protein, known as amino acids. There are about 20 different amino acids. Their order is specified by lengths of mRNA three bases long, called triplet codons. The order of bases in each codon is the code for a particular amino acid (hence the term "genetic code"). The mRNA carries instructions to make a specific protein from a sequence of amino acids.



1 TRANSCRIPTION

In the cell's nucleus, the DNA strands temporarily separate, with one acting as the template for the formation of mRNA. Separate RNA nucleotides with the correct bases lock onto the exposed DNA bases in cross-linked fashion, thereby forming a mirror image of the DNA's information.

2 TRANSLATION

In the cell's cytoplasm, the mRNA attaches to a ribosome. Individual tRNA (transfer ribonucleic acid) molecules have specific amino acids attached. They can slot onto the mRNA only if the order of their bases matches, ensuring they bring the correct amino acid. As the ribosome moves along the mRNA, the tRNAs bring the correct sequence of amino acids, which fit together to construct a protein.

